

KUOPIO, Finland

WMO: 029170

Lat: 63.0083N

Lon: 27.7994E

Elev: 323

StdP: 14.53

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-17.2	-11.3	-22.2	1.6	-16.8	-15.7	2.4	-11.3	23.0	28.1	20.8	28.4	2.3	310	0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.5	79.9	64.9	76.6	63.3	73.4	61.8	67.5	75.4	65.6	73.1	63.8	70.7	7.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.6	92.3	70.6	62.7	86.3	69.1	60.8	80.5	67.6	32.1	75.8	30.6	72.7	29.2	70.8	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.7	17.4	15.6	DB	-21.0	83.8	7.3	4.0	-26.2	86.6	-30.5	88.9	-34.6	91.2	-39.9	94.1	(4)
(5)			WB	-21.3	69.7	7.2	2.6	-26.5	71.5	-30.7	73.0	-34.7	74.5	-40.0	76.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.1	16.4	16.2	24.6	36.3	48.4	58.2	63.9	60.6	50.9	39.5	30.4	22.4
	DBStd	18.78	13.27	13.46	9.17	7.07	8.08	6.66	5.92	5.70	5.89	6.87	8.10	12.25
	HDD50	5158	1041	946	788	411	129	9	0	2	60	328	589	855
	HDD65	9550	1506	1366	1253	860	518	223	92	161	423	789	1039	1320
	CDD50	1188	0	0	0	2	79	254	433	329	87	4	0	0
	CDD65	103	0	0	0	0	2	18	59	24	0	0	0	0
	CDH74	579	0	0	0	0	25	109	336	109	0	0	0	0
	CDH80	87	0	0	0	0	1	13	58	15	0	0	0	0
Wind	WSAvg	7.3	6.8	7.0	7.4	6.7	7.0	7.2	6.5	6.7	7.3	8.1	8.4	8.0
Precipitation	PrecAvg	25.8	2.0	1.6	1.4	1.3	1.9	2.8	3.5	2.8	2.1	2.3	2.0	2.1
	PrecMax	31.5	3.2	2.9	2.7	2.5	3.7	5.2	5.7	6.2	4.0	3.7	4.9	3.4
	PrecMin	21.1	0.8	0.2	0.3	0.4	0.4	0.7	2.1	0.7	0.6	0.6	0.4	0.8
	PrecStd	3.1	0.7	0.7	0.5	0.6	1.0	1.2	1.2	1.3	1.0	0.8	1.1	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.3	39.7	47.4	63.0	77.6	82.1	85.3	81.9	69.0	55.7	48.4	43.7
		MCWB	36.9	36.8	39.8	50.0	58.9	64.4	67.7	65.5	59.7	52.0	46.9	42.4
	2%	DB	35.7	36.7	42.0	55.9	71.8	77.1	81.0	76.9	64.9	53.2	45.4	39.8
		MCWB	34.7	35.3	36.6	46.1	56.3	62.1	66.3	63.7	58.1	50.8	44.0	38.8
	5%	DB	34.0	34.4	38.7	51.7	66.6	73.3	77.8	73.5	62.1	51.3	42.9	37.5
		MCWB	33.4	33.5	34.9	42.8	53.7	60.7	64.8	62.6	56.5	49.3	41.9	36.4
	10%	DB	32.4	32.7	36.0	47.8	62.3	69.7	74.4	70.3	59.4	49.3	40.6	35.4
		MCWB	32.0	32.0	33.3	40.9	51.3	59.0	63.2	61.0	55.1	47.6	39.8	34.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.1	37.5	40.5	52.2	61.5	67.8	70.9	68.4	61.2	53.3	47.2	42.2
		MCDB	37.8	39.0	45.9	59.5	72.9	75.8	79.6	77.2	65.8	54.9	48.2	43.1
	2%	WB	34.8	35.3	37.3	47.6	58.4	65.2	68.5	65.6	59.3	51.4	44.0	38.9
		MCDB	35.3	36.3	40.6	54.6	67.9	72.5	76.7	73.1	63.4	52.9	44.9	39.5
	5%	WB	33.4	33.7	35.4	44.1	55.4	62.6	66.8	64.0	57.6	49.5	41.8	36.4
		MCDB	33.9	34.3	37.7	49.9	64.3	70.3	74.6	71.2	61.3	50.8	42.6	37.1
	10%	WB	32.1	32.1	33.8	41.2	52.5	60.0	65.0	62.3	55.8	47.6	39.6	34.4
		MCDB	32.5	32.7	35.5	46.4	60.7	68.3	72.1	68.4	58.9	48.9	40.3	35.0
Mean Daily Temperature Range	5% DB	MDBR	10.1	11.8	13.8	14.7	15.8	13.9	13.5	12.6	10.3	7.0	6.4	8.5
		MCDBR	8.4	9.8	14.4	21.3	23.3	18.8	17.6	16.8	13.4	8.5	7.6	6.9
		MCWBR	8.2	9.1	10.3	12.6	11.0	8.3	7.1	6.7	7.9	6.8	7.4	6.7
	5% WB	MCDBR	8.3	9.6	11.5	20.0	19.6	16.3	15.7	14.8	12.1	8.0	7.3	6.9
		MCWBR	8.3	9.1	8.9	12.6	10.2	8.4	7.4	7.0	7.9	7.0	7.2	6.7
Clear-Sky Solar Irradiance	taub	0.229	0.286	0.313	0.347	0.345	0.343	0.367	0.350	0.327	0.312	0.246	N/A	
	taud	2.002	2.114	2.184	2.341	2.465	2.494	2.442	2.475	2.528	2.394	2.091	N/A	
	Ebn at Noon	125	202	246	265	277	279	268	263	248	199	117	N/A	
	Edh at Noon	10	21	29	31	30	30	31	27	22	16	9	N/A	
All-Sky Solar Radiation	RadAvg	59	236	718	1156	1518	1669	1589	1213	676	261	65	26	
	RadStd	8	55	86	138	167	106	161	136	76	45	12	3	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.99	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)		+1.03	N/A	N/A	N/A	N/A	N/A	-319	-344	N/A	N/A

Nomenclature: See separate page